

# ALI SHAHIDI

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## SUMMARY

Software Developer focused on Unity/XR systems, C#/.NET/ Python backend engineering, and applied AI. Built ASP.NET Core APIs with JWT and role-based access, delivered multiplayer VR/AR and multimodal AI systems, and used agent/automation workflows to improve delivery speed, reliability, and maintainability.

## PROFESSIONAL EXPERIENCE

**Software Developer** *September 2024 - November 2025*  
**ETHEREAL Research Group** *Calgary, AB*

- Built an accessible VR training system for non-speaking autistic users, improving interaction flow and ethical UX outcomes.
- Engineered backend services to synchronize GPT-driven dialogue, user events, and multimodal sensory feedback in real time.
- Implemented AI behavior analysis that improved video processing speed by 90% and model accuracy by 83%.
- Reduced AI API costs by 98% by redesigning the pipeline to send high-priority metadata instead of raw video streams.

**Software Developer** *September 2024 - March 2025*  
**ELIXR & Illumia** *Calgary, AB*

- Delivered VR/AR solutions across 5 projects using Unity, C#, WebXR, real-time audio, and AI-driven interaction systems.
- Built cockpit and ATC simulators with agent-style LLM orchestration connected to .NET backend APIs.
- Developed RAG pipelines with Llama 3 and Ollama for context-aware querying of domain documents.
- Improved Unity runtime performance through profiling, batching, GPU instancing, and occlusion culling.

**Software Developer - Internship** *February 2024 - August 2024*  
**Telus** *Calgary, AB*

- Led development of a cross-platform multiplayer AR app for iOS and Quest, enabling real-time shared experiences for up to 50 users.
- Engineered NetCode and spatial anchor systems for accurate mapping, animation, and interactive AR workflows.
- Enhanced operational efficiency, reducing transportation costs by 40% through immersive remote instruction.

**Graduate Research & Teaching Assistant - Software Engineer** *January 2021 - April 2024*  
**University of Calgary** *Calgary, AB*

- Supported instruction in software engineering, object-oriented programming, and scalable data analysis.
- Built software prototypes and data workflows for assistive technology research with interdisciplinary collaborators.

## EDUCATION

**University of Calgary** *April 2024*  
Master of Software Engineering (GPA: 4/4.00) *Calgary, Canada*

- Received full graduate funding and an international tuition award; nominated for the Distinguished Master's Thesis Award in STEM.

**University of Tehran** *July 2020*  
B.Sc. in Computer Engineering (GPA: 17.24/20.00) *Tehran, Iran*

- Received Best Undergraduate Project and two Faculty of Engineering Fellowship Awards.

## PROJECTS

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### **Social Networking API** | ASP.NET Core, MVC, REST APIs, MSSQL, Dapper, Entity Framework

- Built a full-stack backend using ASP.NET Core and MSSQL following MVC architecture, exposing RESTful APIs for user management with Swagger documentation.
- Implemented JWT authentication, secure password hashing, and role-based authorization for user/admin workflows.

### **ETHEREAL Research Group - Coffee Shop Simulation** | Unity, C#, .NET, AI, API, VR, SQL

- Developed a high-fidelity VR training environment for neurodivergent users, integrating multimodal sensory feedback and ethical UX design.
- Engineered a .NET backend for real-time state management and asynchronous communication between Unity and external LLM services.
- Integrated GPT dialogue and voice synthesis for adaptive NPC interactions and rubric-style trainee feedback.

### **ETHEREAL Research Group - Multi Modal Behaviour Analysis** | Python, AI, Azure, Unity, VR/AR

- Architected a cross-platform real-time inference system for Meta Quest 3 and mobile tablets, synchronizing multimodal inputs (Vision, Audio, Text).
- Engineered a Python serverless pipeline on Azure for GPU-intensive workloads and high-throughput multimodal processing.
- Optimized ingestion and orchestration to support near real-time AI feedback.

### **ELIXR & Illumia - ATC** | Unity, C#, Network, Optimization, VR, Python, RAG, API

- Engineered a Retrieval-Augmented Generation (RAG) pipeline utilizing Llama 3 and Ollama for real-time querying of domain-specific ATC manuals, with metadata-filtered retrieval, top-k selection, and re-ranking for higher precision under noisy transcripts.
- Developed a low-latency audio/transcription pipeline in Python, achieving 86% command extraction accuracy across diverse accents.

### **Internal Operations Portal (ETHEREAL)** | ASP.NET Core, C#, REST APIs, SQL, DTOs, ViewModels

- Built an internal-use .NET portal for project tracking and workflow visibility, using ASP.NET Core APIs and structured DTO/ViewModel mappings.
- Implemented role-aware endpoints and centralized validation to improve maintainability across modules.

### **Community Sim** | Python, APIs, JSON, Automation

- Developed and maintained Community Sim in Python, integrating API workflows and structured JSON data exchange.
- Built agent/skill-driven automation workflows for setup and validation, improving release consistency and reducing manual overhead.

## SKILLS

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**Languages:** C#, Python, C, C++, JavaScript, SQL

**AI & Machine Learning:** RAG, Llama 3, OpenAI API, agent/skill workflows, multimodal inference, NLP, Azure AI

**XR & Systems:** Unity, Meta Quest SDK, WebXR, NetCode (Multiplayer), Spatial Anchors, AR Foundation

**Backend & Cloud:** ASP.NET Core, MVC, RESTful APIs, Azure, Docker, Micro-ORMs (Dapper)

**Databases:** MS SQL, MySQL, PostgreSQL

**Tools & Workflow:** Git, Linux, LaTeX, Cursor, Claude, Automation, Scrum/Agile, Jira, Confluence